MTH 451/551
Lecture 9

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1 Lagrange Interpolation: Error Estimates

[Textbook reference: §3.4]

We previously saw interpolation of data points (x_i, y_i). We mentioned that a common case we encounter is when $y_i = f(x_i)$ for some function f that we wish to approximate.

> Our question is: **can we quantify how well the polynomial interpolant p approximates the given function f?**

The following theorem summarizes the main error estimate.

Theorem 1 (Lagrange Interpolation Error Theorem)**.** *Suppose that $f \in C^{n+1}[a,b]$, and $p \in \mathbb{P}_n$ interpolates f at the distinct nodes $\{x_0, x_1, \dots, x_n\} \subset [a,b]$. For each $x \in [a,b]$ there is a $\xi(x) \in [a,b]$ for which*

$$f(x) - p(x) = \frac{f^{(n+1)}(\xi(x))}{(n+1)!} \prod_{j=0}^{n} (x - x_j) = \frac{f^{(n+1)}(\xi(x))}{(n+1)!} \ell(x), \tag{1}$$

where $\ell(x)$ is defined by

$$\ell(x) = \prod_{j=0}^{n} (x - x_j) = (x - x_0)(x - x_1) \cdots (x - x_n).$$

Furthermore, $\xi(x)$ may be chosen (for each x) in such a way that $f^{(n+1)}(\xi(x))$ is continuous on $[a,b]$.

Now fix $1 \le k \le n$. There are distinct points $\{\xi_{k,0}, \dots, \xi_{k,n-k}\} \subset (a,b)$ such that, for each $x \in [a,b]$, there is a $\xi_k(x) \in (a,b)$ for which

$$f^{(k)}(x) - p^{(k)}(x) = \frac{f^{(n+1)}(\xi_k(x))}{(n-k+1)!} \prod_{j=0}^{n-k} (x - \xi_{k,j}). \tag{2}$$

Furthermore, $\xi_k(x)$ may be chosen (for each x) in such a way that $f^{(n+1)}(\xi_k(x))$ is continuous on $[a,b]$.

We will prove this theorem later in this lecture.

In the special case that $k = 1$ and $x = x_j$ is one of the interpolation nodes, we can improve upon (2) as follows:

$$f'(x_j) - p'(x_j) = \frac{f^{(n+1)}(\xi(x_j))}{(n+1)!} \prod_{\substack{k=0 \\ k \neq j}}^{n} (x_j - x_k) = \frac{f^{(n+1)}(\xi(x_j))}{\lambda_j (n+1)!},$$

where λ_j are the barycentric weights.

The above results measure the error between f and the interpolating polynomial p *at one specific point.* In order to give some notion of how close one function is to another, it will be useful for us to consider measures of error that consider more than just a single point. The tools that we use to do this are called **norms**, which assign a "size" to functions. As one particularly useful example, we consider the maximum norm.

Definition 1 (Max Norm)**.** *For $f \in C[a,b]$, we define*

$$\|f\|_{C[a,b]} = \max_{x \in [a,b]} |f(x)|.$$

This defines a norm on $C[a,b]$, and can be thought of as measuring the "size" or "length" of f on the closed interval $[a,b]$. For two functions $f, g \in C[a,b]$, $\|f - g\|_{C[a,b]}$ is a natural measure of the "distance" between them.

Example 1. Suppose that $p \in \mathbb{P}_2$ interpolates a function $f \in C^3[-2,1]$ at the nodes $\{-1, 0, 1\}$, and let $\ell(x) = (x+1)x(x-1)$ denote the associated node polynomial. For $x \in [-2,1]$, the error formulas from Theorem 1 are the following:

1. $f(x) - p(x) = \dfrac{f'''(\xi(x))}{6}\ell(x)$ for some $\xi(x)$ between $\min\{x, -1\}$ and $\max\{x, 1\}$;
2. $f'(x) - p'(x) = \dfrac{f'''(\xi_1(x))}{2}(x - \xi_{1,0})(x - \xi_{1,1})$, where $\xi_{1,0} \in (-1, 0)$, $\xi_{1,1} \in (0, 1)$, and $\xi_1(x)$ is (strictly) between $\min\{x, \xi_{1,0}\}$ and $\max\{x, \xi_{1,1}\}$;
3. $f''(x) - p''(x) = f'''(\xi_2(x))(x - \xi_{2,0})$, where $\xi_{2,0} \in (-1, 1)$, and $\xi_2(x)$ is (strictly) between x and $\xi_{2,0}$.

We use the first of these formulas to bound the error $f - p$ in the max norm. If $x \in [-1,1]$, then $\xi(x) \in [-1,1]$ as well, and so $|f'''(\xi(x))| \leq \|f'''\|_{C[-1,1]}$. Therefore,

$$|f(x) - p(x)| \leq \frac{\|f'''\|_{C[-1,1]}}{6}|\ell(x)| \quad \text{for all } x \in [-1,1],$$

and so

$$\|f - p\|_{C[-1,1]} \le \frac{\|f'''\|_{C[-1,1]}}{6}\|\ell\|_{C[-1,1]}.$$

To compute $\|\ell\|_{C[-1,1]}$, note that $\ell(x) = x^3 - x$, so $\ell'(x) = 3x^2 - 1$, which vanishes at $x = \pm 1/\sqrt{3}$. Since $\ell(\pm 1) = 0$ and $\ell(\pm 1/\sqrt{3}) = \mp 2/(3\sqrt{3})$, we have $\|\ell\|_{C[-1,1]} = 2/(3\sqrt{3})$, and so

$$\|f - p\|_{C[-1,1]} \le \frac{\|f'''\|_{C[-1,1]}}{9\sqrt{3}}.$$

On the larger interval $[-2, 1]$, the same argument gives

$$\|f - p\|_{C[-2,1]} \le \frac{\|f'''\|_{C[-2,1]}}{6}\|\ell\|_{C[-2,1]}.$$

The only critical points of ℓ are still $\pm 1/\sqrt{3}$, but now we must also check the endpoint $x = -2$, where $\ell(-2) = (-1)(-2)(-3) = -6$. So $\|\ell\|_{C[-2,1]} = 6$, and

$$\|f - p\|_{C[-2,1]} \le \|f'''\|_{C[-2,1]}.$$

Since $\|f'''\|_{C[-2,1]} \ge \|f'''\|_{C[-1,1]}$, this bound is at least $9\sqrt{3} \approx 15.6$ times larger than the bound on $[-1, 1]$. The reason is that $[-2, -1]$ lies *outside* the nodes, so there we are **extrapolating** rather than interpolating. For $x < -1$, all three factors $x + 1$, x, and $x - 1$ of $\ell(x)$ grow in size as x moves away from the nodes, so $|\ell(x)|$ grows like $|x|^3$. Inside $[-1, 1]$, on the other hand, x is always close to at least one node, which keeps $|\ell(x)|$ small. ♦

We now move on to proving the error estimates. We will use repeatedly Rolle's Theorem.

Theorem 2 (Rolle's Theorem)**.** *Suppose that $f \in C[a, b]$ and f is differentiable on (a, b). If $f(a) = f(b)$, then there is some $c \in (a, b)$ for which $f'(c) = 0$.*

Proof of Theorem 1. We may assume without loss of generality that $x_0 < x_1 < \cdots < x_n$. Set $\xi_{0,j} = x_j$ for all j. We know that $f(x_j) - p(x_j) = 0$.

By Rolle's Theorem applied to $f(x) - p(x)$ on each pair of adjacent points $\xi_{0,j}$, there are points $\xi_{1,j} \in (\xi_{0,j}, \xi_{0,j+1})$ such that $f'(\xi_{1,j}) - p'(\xi_{1,j}) = 0$. We can then apply Rolle's Theorem again, but now to $f'(x) - p'(x)$ on each pair of adjacent points $\xi_{1,j}$, obtaining new points $\xi_{2,j} \in (\xi_{1,j}, \xi_{1,j+1})$ such that $f''(\xi_{2,j}) - p''(\xi_{2,j}) = 0$. Repeating this process, we obtain, for all $1 \le k \le n$

$$\xi_{k,j} \in (\xi_{k-1,j}, \xi_{k-1,j+1}), \quad 0 \le j \le n - k, \qquad \text{such that } f^{(k)}(\xi_{k,j}) - p^{(k)}(\xi_{k,j}) = 0.$$

Now, given k and for fixed $x \in [a, b] \setminus \{\xi_{k,0}, \dots, \xi_{k,n-k}\}$, define $g \in C^{n-k+1}[a, b]$ by

$$g(t) = f^{(k)}(t) - p^{(k)}(t) - \left(f^{(k)}(x) - p^{(k)}(x)\right) \prod_{j=0}^{n-k} \frac{t - \xi_{k,j}}{x - \xi_{k,j}}. \tag{3}$$

Note that

$$g(\xi_{k,j}) = 0 \qquad \text{and} \qquad g(x) = 0.$$

So, $g(t)$ has at least $n-k+2$ roots. Therefore, by repeated application of Rolle's Theorem, there exists some $\xi_k(x) \in (\min(x, \xi_{k,0}), \max(x, \xi_{k,n-k}))$ such that $g^{(n-k+1)}(\xi_k(x)) = 0$. Differentiating both sides of (3) $n-k+1$ times and evaluating at ξ_k yields

$$0 = g^{(n-k+1)}(\xi_k(x)) = f^{(n+1)}(\xi_k(x)) - \left(f^{(k)}(x) - p^{(k)}(x)\right)(n-k+1)! \prod_{j=0}^{n-k} \frac{1}{x - \xi_{k,j}}. \tag{4}$$

To obtain the above, we used that $p^{(n+1)} \equiv 0$ and that $\prod_{j=0}^{n-k}(t - \xi_{k,j})$ is a polynomial of degree $n-k+1$ with leading coefficient 1. Rearranging (4) gives

$$f^{(k)}(x) - p^{(k)}(x) = \frac{f^{(n+1)}}{(n-k+1)!} \prod_{j=0}^{n-k}(x - \xi_{k,j}),$$

which is exactly the conclusion (2). The case of $x = x_{k,j}$ for some j is immediate since both sides are zero. When $k = 0$, we obtain (1).

The proof of the statement that the points ξ_k can be chosen such that $f^{(n+1)}(\xi_k)$ is continuous is omitted here (see the textbook). □